SMS SPAM DETECTION USING DYNAMIC EVIDENCE ROUTING: A TRANSFORMER-BASED APPROACH

U. Jaya Sindhura
(Koneru Lakshmaiah Education Foundation, Hyderabad, Telangana, India ORCID: [optional], [2420030044@klh.edu.in])

K. Ushasri
(Koneru Lakshmaiah Education Foundation, Hyderabad, Telangana, India ORCID: [optional], [2420030376@klh.edu.in])

P. Smitha
(Koneru Lakshmaiah Education Foundation, Hyderabad, Telangana, India ORCID: [optional], [2420030583@klh.edu.in])

Abstract

SMS communication remains an important channel for personal communication, commercial notifications, service updates, and organizational interactions. However, it is also used to distribute
unsolicited advertisements, fraudulent offers, deceptive notifications, and messages intended to obtain sensitive information. The detection of SMS spam is challenging because messages are short, informal, and highly variable in their vocabulary and structure. Conventional machine-learning methods can identify recurring terms and patterns, but they may not fully capture contextual relationships. Transformer-based models offer richer text representations, although their predictions can still be difficult to interpret and may not rely on the same evidence for every message.

This paper presents an SMS spam detection approach based on a Dynamic Evidence Routing Transformer. The implemented system uses a DistilBERT-based representation and combines three evidence pathways: semantic evidence, pattern evidence, and context/entity

evidence. A dynamic routing mechanism assigns contributions to these pathways before the final ham or spam classification. The system also provides prediction confidence information and routing values

that can be examined when analyzing individual messages.
The reported evaluation was conducted on a test set containing 517 SMS messages. The proposed model achieved 99.42% accuracy, 100.00% precision, 95.38% recall, and an F1-score of 97.64%. Its confusion matrix contained 452 true negatives, 62 true positives, three false negatives, and zero false positives. On the same reported evaluation split, TF-IDF with Logistic Regression achieved 96.32% accuracy and an F1-score of 82.88%, while TF-IDF with Linear Support Vector Machine achieved 98.07% accuracy and an F1-score of 91.94%. An ablation study was also conducted to examine the effects of removing individual evidence pathways and replacing dynamic routing with static equal fusion. The findings demonstrate strong classification performance on the evaluated test

set. Further testing on additional datasets and different message sources is necessary to determine how well the model generalizes to real-world SMS traffic.

Key Words: SMS spam detection, text classification, natural language processing, transformer models, DistilBERT, dynamic evidence routing, semantic analysis, pattern detection, machine learning.

Category: Computing methodologies; natural language processing; text classification; machine learning.

1.Introduction

1.1 Background

Short Message Service (SMS) is widely used because it is simple, accessible, and supported by almost all mobile communication systems. It allows individuals to exchange messages without requiring advanced applications or continuous internet connectivity. Businesses and service providers also use SMS to communicate transaction updates, delivery notifications, appointment reminders, promotional offers, and account-related information.

Despite these advantages, SMS is also a channel for unwanted and deceptive communication. Spam messages may advertise products without consent, promote fraudulent prizes, request personal
information, or persuade recipients to visit suspicious websites. Some messages imitate legitimate organizations to create a false impression of trust. These activities can reduce user confidence in mobile communication and expose recipients to privacy and financial risks.

The growing diversity of message content makes spam detection a continuing classification problem. Messages differ in length, writing style, vocabulary, punctuation, abbreviations, and intended purpose. A legitimate message may contain a promotional word, a telephone number, or a reference to a financial transaction. Similarly, a spam message may avoid familiar spam keywords and use ordinary conversational language. Therefore, classification based on one isolated word or a small collection of fixed rules may not be sufficient for every message.

An automated SMS spam detection system attempts to distinguish legitimate messages, commonly referred to as ham, from spam messages. Such a system must learn useful differences between the classes from labelled examples and apply those differences to messages that were not seen during training. Its quality depends on the training data, feature representation, learning procedure, evaluation method, and the degree to which future messages resemble the examples used during development.

Natural language processing provides methods for representing and analyzing textual information. Statistical techniques can identify important words and phrases, while neural language models can learn representations that reflect relationships among words. This research investigates a model that combines multiple forms of evidence to classify SMS messages.

1.2 Evolution of Spam Detection Techniques

Early spam filtering systems often relied on manually defined rules and keyword matching. These systems searched for known suspicious expressions, repeated promotional phrases, unusual formatting, or other
predefined indicators. Their main advantage was simplicity: individual rules could be inspected and understood easily.

However, rule-based systems depend on the quality and coverage of their rules. A new spam message may use unfamiliar wording and avoid known patterns. Conversely, legitimate communication may contain a term that triggers a spam rule. Maintaining a large collection of rules also becomes increasingly difficult as message styles and attack strategies change.

Machine-learning techniques introduced a more flexible approach. Instead of specifying every classification rule manually, a model learns relationships between text features and message labels from a labelled dataset. Methods such as Naive Bayes, Logistic Regression, and Support Vector Machines have been used for text classification. Their performance can be strong when the dataset contains words or phrases that reliably distinguish the target classes.

TF-IDF is a common feature representation in conventional text classification. It assigns numerical weights to terms based on their frequency in a document and their distribution across a collection of documents. These feature vectors can be supplied to a classifier to learn a decision boundary between ham and spam.

Although statistical text classifiers are useful, a bag-of-words representation does not inherently encode contextual relationships in the same manner as a transformer model. A message's interpretation may depend on the relationship between several expressions, the sequence in which they appear, or the context established by the complete sentence.

Transformer-based language models offer an alternative by using attention mechanisms to represent relationships among tokens. BERT introduced bidirectional pre-training for language understanding, while DistilBERT was developed as a smaller model through knowledge distillation (Devlin et al., 2019; Sanh et al., 2019). These models can be adapted to downstream text-classification tasks.
Nevertheless, using a transformer does not automatically solve every spam detection problem. Model predictions depend on the quality and representativeness of the data, and a high confidence score alone does not explain why a message was classified as spam. These considerations motivate the exploration of a model that combines several evidence pathways and exposes their routing contributions for analysis.

1.3 Motivation

SMS messages can provide different types of evidence about their likely class. Some contain recognizable patterns, such as repeated promotional expressions or a direct instruction to claim an unexpected reward. Other messages use less obvious wording, where the relationship between expressions is more informative than a single keyword. Still others contain contextual or entity-related indicators that may contribute to classification.

A model that treats all messages as though they depend on the same type of evidence may not use every useful signal effectively. A more flexible architecture can combine different representations and adjust their relative contributions according to the input.

The proposed Dynamic Evidence Routing Transformer is motivated by this need for input-dependent evidence combination. It uses three pathways: semantic evidence, pattern evidence, and context/entity evidence. The routing mechanism combines their representations before the final classification decision.

The objective is to examine whether this architecture can achieve strong classification performance while providing additional information about the distribution of evidence within the model. The routing weights are useful for describing how the model combines its pathways, but they should not be treated as definitive causal explanations of individual predictions.

1.4 Problem Statement

The primary problem addressed in this research is the accurate classification of SMS messages when useful indicators of spam vary from one message to another.

Several challenges make this task difficult. First, SMS messages are usually short and may contain limited contextual information. Second, spam senders can change their vocabulary, punctuation, and message structure. Third, legitimate messages can share individual words or expressions with spam messages. Fourth, some deceptive messages depend on meaning and context rather than obvious keywords. Finally, classification errors have different consequences: false positives can cause legitimate messages to be blocked, while false negatives allow spam messages to pass through.

The study investigates whether combining semantic, pattern, and context/entity evidence through dynamic routing can provide an effective solution for binary SMS spam classification. The evaluation compares the implemented model with two conventional baselines and examines changes in performance under several ablation conditions.

1.5 Research Objectives

The main objective of this study is to develop and evaluate a transformer-based SMS spam detection system that combines multiple evidence pathways through dynamic routing.

The specific objectives are:

1. To implement an SMS classification model using a DistilBERT-based text representation.
2. To incorporate semantic, pattern, and context/entity evidence pathways within the classification architecture.
3. To combine evidence pathways through the implemented dynamic routing mechanism.
4. To classify messages as ham or spam and report prediction confidence information.
5. To compare the proposed model with TF-IDF-based Logistic Regression and Linear SVM baselines.
6. To evaluate classification performance using accuracy, precision, recall, F1-score, and a confusion matrix.
7. To investigate the effects of removing individual evidence pathways. 8. To examine the reported routing distributions for representative messages.
9. To identify limitations and areas requiring additional evaluation before real-world deployment.

1.6 Research Questions

The study addresses five research questions.

RQ1: How effectively does the Dynamic Evidence Routing Transformer classify SMS messages on the reported test set?

RQ2: How does its performance compare with TF-IDF-based Logistic Regression and Linear SVM when the models are evaluated on the same reported split?

RQ3: How does removing an individual evidence pathway affect the classification metrics?

RQ4: How does the model distribute routing weights across the semantic, pattern, and context/entity pathways for different example messages?

RQ5: What additional validation is required to determine whether the model can generalize to different message sources and changing spam patterns?

These questions separate measured classification performance from broader claims about interpretability and generalization. The reported test results address the first two questions directly. Ablation results and routing outputs provide evidence relevant to the third and fourth questions, while the fifth requires further testing beyond the current evaluation.

1.7 Research Contributions

The study makes several contributions specific to the implemented system and its reported experiments.

First, it presents a multi-pathway SMS classification architecture combining semantic, pattern, and context/entity evidence. Second, it uses dynamic routing to combine the pathway representations rather than relying exclusively on one fixed evidence combination. Third, it evaluates the system against TF-IDF with Logistic Regression and TF-IDF with Linear SVM on the same reported evaluation split. Fourth, it examines pathway-removal variants and static equal fusion to investigate how the model behaves under alternative architectural configurations. Fifth, it reports routing distributions and example-level predictions to provide additional descriptive information about model behavior.

These contributions should be interpreted within the scope of the implementation and evaluation. The study does not claim that transformer models, dynamic routing, or multi-pathway learning are new concepts in general. Its focus is the application and evaluation of this combination in the implemented SMS spam detection system.

1.8 Scope of the Study

The current system focuses on binary text classification. It predicts whether a message is ham or spam based on the learned model. It does not independently verify a sender's identity, inspect the actual destination of a link, detect malware on a device, or establish whether a message violates a particular law or policy.

The reported test set contains 517 messages. Accordingly, the results describe the performance measured on that set and should not be assumed to represent every language, country, network, or future spam strategy. Larger and more diverse evaluations would be necessary to establish broader generalization.

2. Related Work

2.1 Rule-Based SMS Spam Detection

Rule-based spam detection identifies messages using manually specified conditions. These conditions may involve suspicious phrases, promotional expressions, unusual character patterns, telephone numbers, or links. When a message satisfies one or more rules, the system may classify it as spam or assign it a risk score.

A key advantage of this approach is interpretability. A reviewer can inspect the rule that triggered the decision and determine why the message was flagged. Rule-based methods can also be computationally inexpensive, making them suitable for simple filtering applications.

However, their effectiveness depends on the selected rules and the extent to which those rules represent the messages encountered in practice. A spam sender can alter a message to avoid a known pattern, while legitimate communications can accidentally satisfy the same condition. For example, the presence of a promotional word does not necessarily indicate that a message is malicious. Rule-based systems therefore provide a useful starting point but may require extensive maintenance to handle changing language and message formats.

2.2 Statistical and Machine-Learning Approaches

Statistical text-classification methods learn patterns from labelled examples. Instead of manually specifying every possible rule, the system learns a relationship between the input representation and the target label. Naive Bayes, Logistic Regression, and Support Vector Machines are examples of methods commonly applied to text classification.

Naive Bayes uses probability estimates and a conditional-independence assumption to calculate class likelihoods. Its simplicity and efficiency make it useful for text categorization, although the independence assumption does not fully describe the relationships among all words in natural language.

Logistic Regression learns a decision function that maps input features to class probabilities. When paired with TF-IDF features, it can identify terms and combinations of terms associated with spam. Its relative
simplicity makes it useful for establishing a baseline against which more complex systems can be evaluated.

Support Vector Machines learn a separating boundary between classes. A linear SVM can be effective when the text is represented as a high-dimensional sparse vector. Its decision boundary is determined by the training examples and the selected feature representation.

The performance of these methods depends on the dataset and experimental configuration. No classifier should be considered universally superior without a controlled comparison. In this study, Logistic Regression and Linear SVM serve as direct baselines for the proposed model.

2.3 TF-IDF-Based Text Representation

TF-IDF is a statistical feature representation designed to reflect the importance of terms in a document collection. Term frequency measures how often a term occurs within a document, while inverse document frequency reduces the weight of terms that occur across many documents.

A common formulation is:

$$
\operatorname{TFIDF}(t,d)=
\operatorname{TF}(t,d)\times\operatorname{IDF}(t),
$$

where represents a term and represents a document.

The exact calculation depends on the chosen TF-IDF implementation and its settings. Nevertheless, the general principle is to assign higher importance to terms that are informative for particular messages.

For SMS spam detection, TF-IDF can represent recurring words and expressions associated with promotional content, requests, and other
class-specific patterns. The resulting vectors can be processed efficiently by conventional linear classifiers.

A limitation is that conventional TF-IDF features do not inherently represent context in the same way as contextual transformer embeddings. The same word generally receives a feature representation based on its term statistics rather than a contextual representation that changes according to the complete sentence. This limitation motivates evaluating a transformer-based approach alongside TF-IDF baselines.

2.4 Transformer-Based Text Classification

Transformer architectures use attention mechanisms to model relationships between tokens in a sequence. Unlike representations based only on independent term counts, contextual representations can encode information about how words are used in relation to surrounding text.

BERT introduced bidirectional pre-training for language understanding and demonstrated the usefulness of pre-trained representations for downstream NLP tasks (Devlin et al., 2019). The model can be adapted to classification by attaching a suitable output component and training it on labelled examples.

DistilBERT applies knowledge distillation to create a smaller transformer model while retaining substantial language-modeling capability (Sanh et al., 2019). This can be beneficial where computational resources and inference time are important considerations.

In SMS classification, transformer-based representations may help distinguish messages whose interpretation depends on the relationship among multiple expressions. However, the quality of a transformer classifier still depends on the training data, model configuration, and evaluation protocol. It is also possible for a transformer model to make errors or learn patterns that do not generalize to future messages.

The proposed system uses a DistilBERT-based representation as a component of a wider evidence-routing architecture. The transformer
supplies textual representations, while the evidence pathways and routing mechanism determine how the information is combined for classification.

2.5 Multi-Pathway Evidence Combination

Multi-pathway architectures aim to represent different aspects of an input through separate components. For text classification, these components may emphasize different features or representations before their outputs are combined.

Evidence can be combined using concatenation, averaging, weighted sums, or learned gating mechanisms. Fixed fusion applies the same combination rule across inputs. Dynamic fusion allows the relative contributions of components to vary according to the input or its learned representation.

Dynamic evidence combination may be useful when messages differ in the signals they contain. For instance, one message may contain strong recurring patterns, while another may require more semantic interpretation. A routing mechanism provides a way to adjust the contributions of these evidence sources.

However, the use of multiple pathways does not guarantee improved performance. A pathway may provide redundant information, and a learned routing mechanism may favor one pathway without producing measurable gains. Therefore, direct evaluation and ablation experiments are necessary to assess whether the added components contribute useful information.

2.6 Research Gap

The present study investigates a particular implementation that combines semantic, pattern, and context/entity evidence pathways using dynamic routing for SMS spam classification. The evaluation considers conventional baselines, pathway-removal variants, static equal fusion, and routing distributions for representative messages.
The study's central question is whether this combination produces useful classification performance on the selected test set and whether the routing outputs provide additional descriptive information about model behavior. The experiments provide initial evidence for the implemented system, but they do not establish universal superiority over existing spam filters or demonstrate robustness across all message sources.

2.7 Related-Work Summary

Approach	Main strength	Main limitation
Rule-based filtering	Directly inspectable rules	Requires maintenance and may miss changed wording
TF-IDF with Logistic Regression	Efficient statistical baseline	Limited inherent contextual representation

TF-IDF with Linear SVM	Effective linear classification for sparse text	Depends on selected features and training data
Transformer-based classification	Contextual text representations	Requires training and careful evaluation
Dynamic Evidence Routing Transformer	Combines multiple evidence pathways with input-dependent routing	More complex and requires further validation

This table summarizes general characteristics of the method families. It is not a claim that every implementation of a particular method has the same performance.

3. Problem Definition and System Overview

3.1 Formal Problem Definition

SMS spam detection can be defined as a supervised binary classification problem. The model receives a text message and predicts whether it belongs to the ham or spam class. Let the labelled dataset be represented as

$$D=\{(x_i,y_i)\}_{i=1}^{N},$$

where denotes the text of message , denotes its ground-truth class,

and represents the number of labelled examples.

The target label belongs to the set

$$y_i\in\{\text{ham},\text{spam}\}.$$

The objective is to learn a classification function, parameterized by ,

that maps an input message to a predicted label:

$$\hat{y}_i=f_\theta(x_i).$$

A probabilistic classifier may additionally estimate the likelihood of each class:

$$

P(y\mid x_i)=
[P(\text{ham}\mid x_i),P(\text{spam}\mid x_i)].
\]
The predicted class is selected according to the decision rule implemented by the model. The confidence associated with a prediction can help describe the model's output, but confidence alone does not establish that the prediction is correct.

The classification task involves two important types of error. A false positive occurs when a legitimate message is predicted as spam. A false negative occurs when a spam message is predicted as ham. A useful model should minimize these errors while maintaining reliable performance across the messages it is expected to process.

3.2 Functional Requirements

The proposed system is designed to perform several connected functions. First, it must accept an SMS message in text form. Second, it must transform the text into the numerical input representation expected by the trained model. Third, it must process the message through the model's evidence pathways and routing mechanism. Finally, it must return the predicted class and any supported confidence or routing information.

The system should also support evaluation on labelled test data. During evaluation, predictions are compared with the true labels to calculate accuracy, precision, recall, F1-score, and confusion-matrix values.

These functions form the basis of the experimental workflow. The classifier itself is responsible for predicting the class, while the evaluation component measures its performance. Separating these functions makes it easier to distinguish a prediction interface from the experiments used to assess the model.

3.3 System Workflow

The proposed workflow consists of the following stages.

Message input:An SMS message is provided to the classification system. The message may originate from the evaluation dataset or from a user interacting with the application.

Tokenization: The text is converted into token identifiers using the tokenizer associated with the transformer model. An attention mask is generated as required by the model.

Representation generation: The tokenized message is processed by the DistilBERT-based component to obtain a representation of the input text.

Evidence processing: The model uses semantic, pattern, and context/entity pathways to produce representations intended to capture different forms of evidence.
Dynamic routing: The routing component determines the relative contributions of the evidence pathways according to the implemented architecture.

Classification: The combined evidence is passed to the classification component, which generates the output used to distinguish ham from spam.

Result presentation: The system returns the predicted class and available confidence information. Routing values can also be examined to describe the contribution assigned to each pathway.

This workflow provides a logical description of the system. The precise order of internal operations and the mathematical form of each component should match the model implementation.

3.4 System Architecture

The system can be described through four functional layers.

The first layer is the input and preprocessing layer. It receives the SMS text and converts it into the input format required by the transformer model. Consistent preprocessing is important because training and inference should use compatible tokenization and input preparation.

The second layer is the text-representation layer. It uses the DistilBERT-based component to obtain contextual text features. These features provide the information processed by the subsequent evidence pathways.

The third layer is the evidence-routing layer. It includes the semantic, pattern, and context/entity pathways and the mechanism that combines their representations. This layer is the central architectural focus of the proposed system.

The fourth layer is the prediction and evaluation layer. It produces the class output and supports the calculation of evaluation metrics. When
the model exposes routing values, these can be recorded for further analysis.

3.5 Input and Output Representation

The input to the system is an SMS message represented as a string of text. Tokenization transforms this string into token identifiers. Depending on the tokenizer configuration, the resulting sequence may be truncated to a maximum length or padded to a fixed length.

The model's primary output is a binary classification result. The ham class indicates that the model predicts a legitimate message, whereas the spam class indicates that the model predicts an unwanted or malicious message according to the learned labels.

Additional outputs may include confidence information and pathway routing values. These outputs are useful for inspecting individual predictions, but their interpretation requires care. A routing weight describes how the model combines internal evidence under its routing mechanism; it does not prove that a particular word caused the classification.

3.6 Design Considerations

The architecture is designed to combine multiple evidence sources without assuming that every message depends on the same signal. The semantic pathway is intended to support meaning-related representations, the pattern pathway to support recurring textual patterns, and the context/entity pathway to provide additional contextual information.

The system also needs to be evaluated using more than one performance measure. Accuracy summarizes the proportion of correct predictions, while precision and recall describe different aspects of spam identification. The F1-score combines precision and recall, and the confusion matrix reveals the number of each type of error.

Finally, the system should be evaluated within a clearly stated scope. A model tested on one collection of SMS messages may encounter
different vocabulary and attack patterns in another collection. Results must therefore be presented with the dataset and evaluation conditions clearly identified.

4. Proposed Dynamic Evidence Routing Methodology 4.1 Overview of the Proposed Model

The proposed Dynamic Evidence Routing Transformer combines a transformer-based text representation with three evidence pathways. The pathways are designed to represent semantic, pattern, and context/entity information. Their outputs are combined through the routing mechanism before the final classification is produced.

The main motivation is that different messages may contain different types of useful evidence. A message with recognizable promotional wording may contain strong pattern-related signals. A more indirect message may depend on the relationship among several expressions. Contextual indicators may also contribute information that is not captured adequately by one evidence type alone.

The architecture aims to provide flexibility in how the evidence is combined. Rather than assuming a fixed contribution for every pathway, the routing mechanism allows the relative contributions to depend on the input, according to the implemented model.

4.2 DistilBERT-Based Representation

DistilBERT is a compact transformer language model developed through knowledge distillation from a larger teacher model (Sanh et al., 2019). It retains a transformer-based architecture while reducing model size and computational requirements compared with its larger counterpart.

The input message is tokenized into a sequence of token identifiers:

\[
X=(x_1,x_2,\ldots,x_T),
\]
where is the effective sequence length. The model also receives an attention mask which identifies the positions that should be treated as input tokens rather than padding.

The text representation can be expressed as

\[
H=\operatorname{DistilBERT}(X,A),
\]

where denotes the representation produced by the transformer.

The representation provides the input to the evidence-processing components. The transformer can represent token relationships in context, but the quality of the resulting classification still depends on the training process and the quality of the labelled data.

The exact pooling operation, hidden dimensions, and projection layers should be

described from the implemented code. These details should not be assumed solely from the use of DistilBERT.

4.3 Semantic Evidence Pathway

The semantic pathway is intended to represent information associated with the meaning of the SMS message. In spam detection, meaning often depends on the relationship among words rather than the presence of an isolated expression.

For example, a message that claims a recipient has won an unexpected reward and requests an action to claim it may contain a combination of signals that is more informative than any individual word. A legitimate message can also contain reward-related or promotional terms, so the overall interpretation remains important.

Let the semantic pathway representation be written as

$$
z_s=f_s(H),
$$

where is the transformer representation and denotes the semantic pathway.

The resulting representation is used as one source of evidence for the final classifier. Its contribution is determined by the routing mechanism, rather than being assumed to have equal importance for every message.

4.4 Pattern Evidence Pathway

The pattern pathway is designed to capture information associated with recurring lexical or structural characteristics. SMS spam messages may share common expressions or patterns, while legitimate messages may contain different combinations of words and sentence structures.

Let the output of the pattern pathway be represented as

$$
z_p=f_p(H),
$$

where denotes the implemented pattern-processing component.

Pattern evidence can be useful when the message contains recognizable signals. However, the same expression can appear in both ham and spam. A reliable classifier must therefore learn from the combination of features rather than treat any single term as a definitive indication of spam.

The final paper should describe the pathway according to its actual implementation. If it processes learned transformer features, it should not be described as a separate handcrafted keyword system unless such a system is explicitly implemented.

4.5 Context/Entity Evidence Pathway

The context/entity pathway provides an additional source of evidence related to contextual or entity-associated information. Depending on the
implementation, this may include information represented in relation to entities or other contextual characteristics of the message.

Its output can be denoted as

$$
z_c=f_c(H),
$$

Where represents the implemented context/entity pathway.

Context-related information can be useful because a message's interpretation depends partly on the surrounding text. Nevertheless, the presence of an organization name, telephone number, or other entity does not independently establish that a message is malicious. Legitimate service notifications can contain these elements as well.

The role of this pathway is to provide an additional representation that can be combined with the other evidence sources. Claims about a separate named-entity recognition system, external databases, or manually extracted entity features should be included only if those components exist in the project implementation.

4.6 Dynamic Routing Mechanism

The routing mechanism determines how the pathway representations contribute to the combined representation. Let the routing weights be represented as

$$
\alpha_s,\quad\alpha_p,\quad\alpha_c,
$$

where the subscripts denote semantic, pattern, and context/entity evidence.

For a normalized weighted-fusion implementation, the weights satisfy

$$
\alpha_s+\alpha_p+\alpha_c=1
$$

and

$$
\alpha_s,\alpha_p,\alpha_c\geq 0.
$$

The combined representation can be expressed as

$$
z=\alpha_s z_s+\alpha_p z_p+\alpha_c z_c.
$$

This equation describes normalized weighted fusion. It should be treated as the exact mathematical operation only if it matches the implemented routing and fusion code. If the model uses a different fusion operation, the equation must be adjusted accordingly.

The main idea is that the routing weights can vary from message to message. This makes the combination input-dependent rather than fixed. A message may receive a relatively large contribution from the pattern pathway, while another may receive more contribution from semantic or context/entity evidence.

Dynamic routing does not necessarily mean that the system chooses only one pathway. Several pathways may contribute to the final representation at the same time. The weights describe the combination used by the model and should not automatically be interpreted as calibrated probabilities or direct measures of causal importance.

4.7 Evidence Fusion and Final Classification

After the pathway representations are combined, the classification component produces scores for the ham and spam classes. Let the
combined representation be, and let the classification function be . The output scores can be represented as

$$
o=g(z).
$$

If the implementation uses a softmax output layer, the probability associated with class is

$$
P(y=k\mid x)=
\frac{\exp(o_k)}
{\sum_j\exp(o_j)}.
$$

The model's decision rule uses its output scores to select the predicted class. The system may also display a confidence score associated with the prediction.

The confidence score describes the model's output rather than guaranteeing correctness. A model can be highly confident and still make a mistake, particularly when the input differs from the examples represented in its training data. Confidence calibration and performance under distribution shifts would require separate experiments.

4.8 Training Objective

Supervised classification requires a learning objective that measures the difference between predicted outputs and target labels. Cross-entropy is a common objective for classification models.

For a two-class problem, the loss for example can be expressed as

$$
\mathcal{L}_i=

-\sum_{k=1}^{2}y_{i,k}\log(p_{i,k}),
\]
where is the target indicator for class , and is the predicted

probability for that class.

For a dataset containing training examples, the average loss is

\[
\mathcal{L}=
-\frac{1}{N}
\sum_{i=1}^{N}
\sum_{k=1}^{2}
y_{i,k}\log(p_{i,k}).
\]

This is the standard cross-entropy formulation. It should be presented as the actual objective used by the project only after checking the training code. If class weighting, a different loss, or additional regularization is used, those details should also be documented.

During training, the model parameters are adjusted to reduce the selected loss. The trained parameters are saved in a checkpoint that can later be loaded for inference. The optimizer, learning rate, batch size, number of epochs, and checkpoint-selection procedure must be taken from the actual training configuration.

4.9 Inference Procedure

During inference, the trained model receives a new SMS message and applies the preprocessing and tokenization required by the model. The resulting token identifiers and attention mask are passed through the forward computation.

The transformer produces a representation, the evidence pathways generate their representations, and the routing mechanism combines the available evidence. The classification component then generates the output used to predict ham or spam.
The implementation can return the predicted label and available confidence information. If routing values are exposed, they can be recorded for later analysis. This allows the researcher to examine both the final classification and the relative contribution of the evidence pathways.

The inference procedure should use the same tokenizer and compatible preprocessing configuration as the trained model. Inconsistent input preparation can cause prediction errors even when the checkpoint itself is valid.

4.10 Algorithmic Description

The general classification workflow is summarized below. **Input:** SMS

text and trained model parameters θ.

^

Output: Predicted class , confidence information, and available routing values.

1. Receive the SMS text.

2. Tokenize the message using the model-associated tokenizer. 3. Generate the attention mask.
4. Compute the transformer representation.
5. Generate semantic, pattern, and context/entity pathway representations.
6. Determine the routing weights using the implemented routing mechanism.
7. Combine the pathway representations according to the implemented fusion operation.
8. Compute the classification scores.
9. Apply the model's decision rule to obtain the predicted class. 10. Return the prediction and available confidence and routing information.

This procedure summarizes the functional behavior of the model without assuming unverified implementation details. For a final reproducible paper, the pseudocode should be aligned with the exact operations in the project's model and inference files.

4.11 Methodological Considerations

The proposed architecture introduces several design considerations. The first is whether each pathway provides useful information beyond the others. The second is whether the routing mechanism improves classification compared with a fixed combination. The third is whether the model maintains performance when tested on messages that differ from its training examples.

The ablation study addresses the first two considerations by examining variants with individual pathways removed and a static equal-fusion configuration. The baseline comparison evaluates the complete system against conventional TF-IDF classifiers.

These experiments provide evidence about performance under the reported evaluation conditions. They do not establish that the routing mechanism will improve every dataset or that the routing weights fully explain the model's decisions. Further evaluation is required to investigate generalization and reliability under different conditions.

5. Implementation and Experimental Setup

5.1 Implementation Environment

The proposed SMS spam detection system was implemented using Python and deep learning libraries. The model uses a DistilBERT-based text representation component together with multiple evidence-processing pathways and a dynamic evidence routing mechanism. The implementation is organized into separate components for model definition, training, evaluation, and application-level prediction. This modular structure makes it easier to test individual components and compare the complete model with alternative configurations.

PyTorch is used for model implementation and tensor operations, while the Transformers library provides the tokenizer and transformer-related functionality. The trained model parameters are stored in a checkpoint file and loaded during inference. A Flask-based web application provides an interface through which users can enter an SMS message and receive a predicted class. The system returns either ham or spam, together with a confidence score and routing information where available.

The separation between training and inference allows the same saved model to be evaluated independently of the web interface. This also makes it possible to investigate

the effects of changes to the routing mechanism without redesigning the complete application.

5.2 Dataset and Data Preparation

The model was evaluated on a test set containing 517 SMS messages. Each message belongs to one of two categories: legitimate messages, referred to as ham, and unwanted or potentially fraudulent messages, referred to as spam. The classification task is therefore a binary text-classification problem.

Before processing, each SMS message is converted into a token sequence suitable for the transformer model. The tokenizer maps words and subword units to numerical token identifiers. An attention mask indicates which positions contain meaningful tokens and which positions are padding. These representations are then passed to the model for classification.

The same reported test set was used for evaluating the proposed model and the two traditional machine-learning baselines. Using a common test set makes the comparison more informative because all three models are evaluated on the same collection of messages.

The dataset name, original source, training-validation-test proportions, random seed, and detailed preprocessing rules should be reported from the actual project configuration. These details should not be assumed when they have not been verified.

Table 1. Summary of the reported test set

Parameter	Description
Task	Binary SMS classification
Classes	Ham and spam
Test samples	517
Proposed model	DistilBERT-based Dynamic Evidence Routing
Evaluation	Classification metrics and confusion matrix

5.3 Proposed Model Configuration

The proposed model, DynamicEvidenceSMSModel, combines transformer-based text representations with three evidence pathways: semantic, pattern, and context/entity evidence. The semantic pathway is intended to capture contextual meaning, the pattern pathway to identify message-level indicators, and the context/entity pathway to represent relevant entities and contextual information.

The dynamic evidence routing mechanism combines information from these pathways to produce the final classification. Rather than assuming that every pathway contributes equally to every message, the model uses routing weights to represent their relative contributions. The resulting classification is based on the evidence available for the individual message.

The model was evaluated using its saved checkpoint. This ensures that the reported predictions correspond to the trained model rather than an untrained initialization. Exact hyperparameters, including learning rate,
batch size, number of epochs, optimizer, and loss function, should be included only after checking the training script or configuration file.

5.4 Baseline Models

Two conventional machine-learning baselines were evaluated alongside the proposed model.

The first baseline combines Term Frequency–Inverse Document Frequency (TF-IDF) text representation with Logistic Regression. TF-IDF represents messages according to the importance of their terms, while Logistic Regression estimates the probability of membership in the spam class. This combination is commonly used as a lightweight text-classification baseline.

The second baseline combines TF-IDF with a Linear Support Vector Machine (Linear SVM). The classifier learns a separating boundary between legitimate and spam messages using their vector representations. Linear SVM is a useful comparison because it can perform well in high-dimensional text-classification tasks without requiring a transformer-based architecture.

Both baselines provide reference points for evaluating whether the proposed transformer-based model improves the classification of spam messages under the reported experimental setup.

5.5 Experimental Procedure

The evaluation procedure consisted of loading the trained model, processing the test messages, obtaining predictions, and comparing the predictions with the reference labels. The evaluation metrics were then calculated from the predicted and actual classes.

The proposed model was compared with TF-IDF plus Logistic Regression and TF-IDF plus Linear SVM. In addition to overall accuracy, precision, recall, and F1-score were examined to understand the model's performance on the spam class. The confusion matrix was also used to identify false-positive and false-negative predictions.

This evaluation provides both an overall view of classification quality and a more detailed understanding of the types of errors made by the model.

6. Evaluation Metrics

The proposed system was evaluated using accuracy, precision, recall, and F1-score. These metrics provide complementary information about the performance of a binary classifier. Accuracy measures the proportion of all correctly classified messages, whereas precision, recall, and F1-score focus more directly on positive-class predictions.

For the following definitions, spam is treated as the positive class. A true positive (TP) is a spam message correctly classified as spam. A true negative (TN) is a legitimate message correctly classified as ham. A false positive (FP) is a legitimate message incorrectly classified as spam, while a false negative (FN) is a spam message incorrectly classified as ham.

6.1 Accuracy

Accuracy measures the proportion of messages classified correctly across both classes.

$$\text{Accuracy}=\frac{TP+TN}{TP+TN+FP+FN}$$

A high accuracy indicates that the model correctly classifies a large proportion of the test messages. However, accuracy alone may not adequately describe spam-detection performance when the classes are imbalanced.

6.2 Precision

Precision measures the proportion of messages predicted as spam that are actually spam.

$$\text{Precision}=\frac{TP}{TP+FP}$$

High precision is important because legitimate messages incorrectly labelled as spam may be hidden, ignored, or placed in a spam folder. Precision therefore provides information about the reliability of positive spam predictions.

6.3 Recall

Recall measures the proportion of actual spam messages correctly detected by the model.

$$\text{Recall}=\frac{TP}{TP+FN}$$

Recall is especially important for spam filtering because a false negative allows an unwanted message to be classified as legitimate. A model with high recall detects a greater proportion of the spam messages in the test set.

6.4 F1-Score

The F1-score is the harmonic mean of precision and recall.

$$F_1=2\frac{\text{Precision}\times\text{Recall}}{\text{Precision}+\text{Recall}}$$

This metric balances precision and recall. It is useful when both the reliability of spam predictions and the ability to detect spam are important.

6.5 Confusion Matrix

The confusion matrix provides a count of correct and incorrect predictions for each class. Unlike a single aggregate score, it reveals the types of errors made by the classifier.

For the proposed model, the reported confusion matrix contains 452 true negatives, zero false positives, three false negatives, and 62 true positives. Thus, no legitimate messages in the test set were incorrectly classified as spam, while three spam messages were missed.

7. Results and Performance Analysis

7.1 Overall Classification Performance

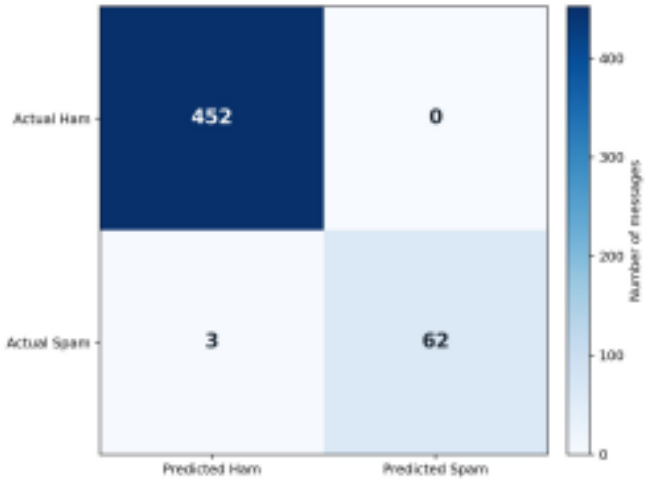
The proposed Dynamic Evidence Routing model achieved an accuracy of 99.42%, precision of 100.00%, recall of 95.38%, and F1-score of 97.64% on the reported test set of 517 messages.

The accuracy indicates that the model classified almost all test messages correctly. Its precision of 100.00% indicates that every message predicted as spam in this test set was actually labelled spam. The recall of 95.38% shows that the model detected most of the actual spam messages, although three spam messages were classified as ham.

The F1-score of 97.64% reflects the balance between the model's precision and recall. Together, these metrics indicate strong performance on this particular test set. They should not, however, be interpreted as proof that the model will achieve the same results on different datasets or in real-world deployment.

Table 2. Confusion matrix of the proposed model

Actual class	Predicted Ham	Predicted Spam
Ham	452	0
Spam	3	62



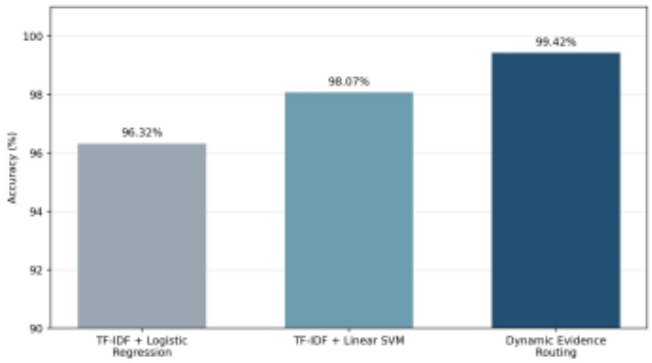
The confusion matrix highlights an important characteristic of the model: it made no false-positive errors in the reported evaluation, but it missed
three spam messages. This distinction is useful because the consequences of these errors differ. False positives can interfere with legitimate communication, while false negatives allow spam messages to pass through the filter.

7.2 Comparison with Baseline Models

The proposed model was compared with TF-IDF plus Logistic Regression and TF-IDF plus Linear SVM using the reported common test set.

Table 3. Performance comparison of evaluated models

Model	Accuracy	Recall	F1-score
TF-IDF + Logistic Regression	96.32%	100.00%	82.88%
TF-IDF + Linear SVM	98.07%	87.69%	91.94%
Proposed Dynamic Evidence Routing	99.42%	100.00%	97.64%



The Logistic Regression baseline achieved 96.32% accuracy and 70.77% recall. Although its precision was 100.00%, its lower recall indicates that it missed a relatively large proportion of the spam messages in this test set.

The Linear SVM baseline improved on Logistic Regression in accuracy, recall, and F1-score. It achieved 98.07% accuracy and 91.94% F1-score. However, its reported precision was lower than that of the proposed model.

The proposed model achieved the highest reported accuracy, recall, and F1-score among these three evaluated configurations. Compared with Linear SVM, its recall was 7.69 percentage points higher, while its F1-score was 5.70 percentage points higher. Its precision was also higher by 3.39 percentage points.

These results support the conclusion that the proposed model performed better than the two tested baselines on this evaluation. They do not establish that it outperforms all existing SMS spam detection methods. Broader conclusions require testing across additional datasets and evaluation conditions.

7.3 Interpretation of False Positives and False Negatives The absence of false positives is a positive result because no legitimate message in the reported test set was

incorrectly labelled as spam. This may reduce the risk of legitimate communication being filtered unnecessarily.

The three false negatives indicate that the model did not detect every spam message. These cases may contain less obvious spam indicators, unusual wording, or contextual patterns that differ from those represented in the training data. The exact reasons cannot be determined without examining the individual misclassified messages.

Future evaluation should therefore examine false-negative examples separately. Such analysis may reveal whether the missed messages
contain misleading wording, previously unseen expressions, or other characteristics that challenge the current model.

7.4 Summary of Findings

The reported results show that the Dynamic Evidence Routing model achieved 99.42% accuracy and a 97.64% F1-score. It also achieved 100.00% precision, with three false negatives and no false positives in the 517-message test set.

The baseline comparison suggests that the proposed configuration improved spam recall over both tested conventional machine-learning models. The results provide evidence for the usefulness of the implemented approach within the evaluated setting, while further testing is needed to assess its generalization to other datasets.

8. Ablation and Routing Analysis

8.1 Purpose of the Ablation Study

An ablation study evaluates how the performance of a model changes when individual components are removed or when the model's combination strategy is modified. It helps investigate whether the complete architecture provides measurable benefits over simplified configurations.

In the proposed SMS spam detection system, the main components are the semantic expert, pattern expert, context/entity expert, and dynamic evidence routing mechanism. The ablation experiments compare the complete model against configurations that exclude one expert or replace dynamic routing with static equal fusion.

The purpose of this analysis is to examine the contribution of each component under the reported test conditions. An ablation result should be interpreted carefully because a component may provide useful information without producing a measurable change in every evaluation metric or on every test set.

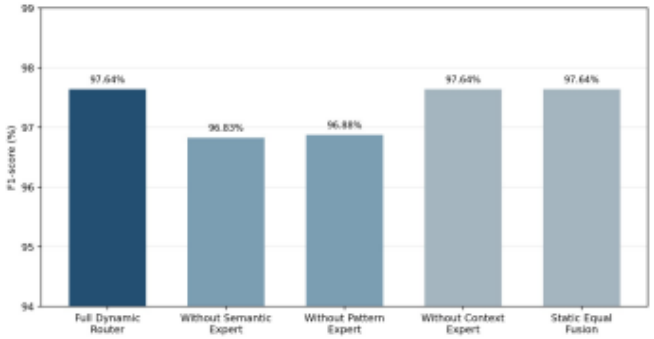
8.2 Ablation Results

The ablation experiments produced the results shown in Table 4. The values are presented as percentages for easier comparison.

Table 4. Ablation study of the proposed model

Configuration	Accuracy	Precision	Recall	F1-score

Full Dynamic Router	99.42 %	100.00%	95.38 %	97.64 %
Without Semantic Expert	99.23 %	100.00%	93.85 %	96.83 %
Without Pattern Expert	99.23 %	98.41 %	95.38 %	96.88 %
Without Context Expert	99.42 %	100.00%	95.38 %	97.64 %
Static Equal Fusion	99.42 %	100.00%	95.38 %	97.64 %



The complete Dynamic Router achieved the highest F1-score among the configurations shown. Removing the semantic expert reduced recall from 95.38% to 93.85%, suggesting that semantic evidence contributed to detecting some spam messages in the evaluated set. Removing the pattern expert reduced precision from 100.00% to 98.41%, while recall remained at 95.38%. This indicates that the pattern pathway may have contributed to the reliability of spam predictions in this evaluation.

The configuration without the context expert produced the same reported metrics as the full model. Static equal fusion also produced the same reported metrics as the complete model. These results mean that no difference was observed in the reported metrics for those configurations on this particular test set.

However, identical aggregate metrics do not prove that the context expert or dynamic routing is unnecessary. Different configurations can produce different internal representations or routing decisions while still yielding the same final predictions.

Additional experiments, larger datasets, and repeated runs would be required to determine whether these components provide consistent benefits.

8.3 Contribution of the Semantic Expert

The semantic expert is intended to capture contextual meaning that may not be fully represented by individual keywords or simple message patterns. For example, a spam message may use indirect language to encourage a recipient to claim a reward or respond to an offer. The meaning of the complete message may be more informative than the presence of a single suspicious word.

The ablation results show that removing the semantic expert reduced recall by approximately 1.54 percentage points and reduced the F1-score by approximately 0.81 percentage points. This suggests that semantic information was useful for the evaluated configuration.

Nevertheless, the results do not establish that semantic evidence is equally important for every message. The contribution of the semantic
expert may depend on the wording, context, and characteristics of individual SMS messages.

8.4 Contribution of the Pattern Expert

The pattern expert is designed to identify message-level indicators associated with spam, such as suspicious phrases, promotional wording, urgency, or requests to claim an offer. Such indicators can be useful when spam messages contain recognizable textual patterns.

When the pattern expert was removed, precision decreased from 100.00% to 98.41%, while recall remained unchanged. This indicates that the pattern expert may have helped distinguish spam-related evidence from legitimate message content in the reported evaluation.

However, the ablation results alone do not identify which specific patterns were responsible for the difference. Further error analysis would be necessary to determine the types of messages that benefit most from this pathway.

8.5 Contribution of the Context/Entity Expert

The context/entity expert is intended to represent contextual or entity-related evidence relevant to message classification. Such information may help distinguish the significance of a phrase according to the surrounding message content.

In the reported ablation experiment, removing this expert did not change accuracy, precision, recall, or F1-score. This means that the full model and the configuration without this expert achieved the same aggregate performance on the test set.

This result should be treated as an observation rather than a general conclusion about the value of contextual evidence. The expert may have limited influence on this particular dataset, or its information may overlap with evidence captured by other pathways. Further experiments should

examine its contribution across different datasets and more challenging messages.

8.6 Dynamic Routing Versus Static Equal Fusion

Dynamic evidence routing is intended to adapt the relative contribution of the evidence pathways to the input message. In contrast, static equal fusion combines the pathways using a fixed equal-weight strategy.

The reported results show that the static equal-fusion configuration achieved the same classification metrics as the full dynamic router. Therefore, the current experiment does not demonstrate an improvement in aggregate classification performance from dynamic routing over static equal fusion on this test set.

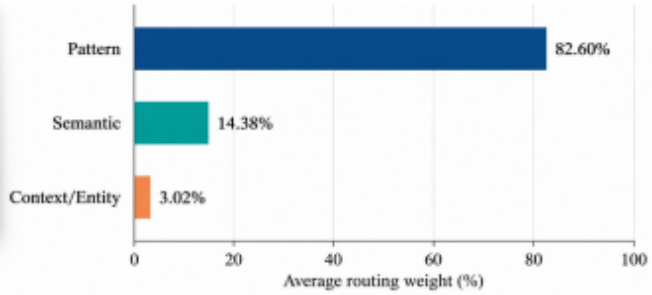
The distinction between the two approaches may still be relevant to how evidence is allocated internally. However, a claim that dynamic routing is more accurate than equal fusion would require evidence of a measurable difference across suitable experiments. Future evaluation should compare prediction-level differences, routing distributions, computational costs, and performance across multiple test sets.

8.7 Average Evidence Routing Distribution

The model's reported average routing distribution was 14.38% for semantic evidence, 82.60% for pattern evidence, and 3.02% for context/entity evidence.

Figure 6. Average evidence routing distribution

Reported average routing weights



The average distribution indicates that the pattern pathway received the largest share of routing weight across the evaluated messages. Semantic evidence received a smaller share, while context/entity evidence received the smallest average share.

This distribution describes the model's reported routing behaviour, but it should not be interpreted as proof that pattern evidence is inherently more important than semantic evidence. The weights depend on the learned model, dataset, architecture, and implementation. A smaller average weight also does not necessarily mean that a pathway has little influence on particular messages.

8.8 Message-Level Routing Behaviour

The routing weights differed between the two example messages examined during testing.

Table 5. Example message-level routing distributions

Message type	Semantic	Pattern	Context/ Entity
Example spam message	74.34%	7.81%	17.85%
Example legitimate message	7.40%	89.52%	3.08%

The example spam message, “Congratulations! You have won a free prize. Call now to claim,” received a spam prediction with a reported confidence of 96.69%. Its routing distribution assigned 74.34% to semantic evidence, 7.81% to pattern evidence, and 17.85% to context/entity evidence.

The example legitimate message, “Hey, are we still meeting for lunch today?”, received a ham prediction with a reported confidence of
98.99%. Its routing distribution assigned 7.40% to semantic evidence, 89.52% to pattern evidence, and 3.08% to context/entity evidence.

These examples illustrate that the model's routing weights can differ across inputs. The spam example relied more heavily on semantic evidence, whereas the legitimate example assigned most of its routing weight to pattern evidence.

The examples are illustrative rather than conclusive. Two messages are insufficient to establish a general rule about how the model routes all spam and legitimate messages. A larger-scale analysis of routing weights grouped by class would provide stronger evidence of systematic differences.

8.9 Summary of Routing Analysis

The ablation and routing experiments provide two types of information. First, the ablation results show how classification metrics changed when components were removed or the fusion strategy was modified. Second, the routing distributions illustrate how the model allocated evidence for individual messages and across the evaluated set.

The full model achieved the highest F1-score among the reported ablation configurations, although static equal fusion and removal of the context/entity expert produced the same aggregate metrics as the complete model. The routing analysis also showed substantial variation between the two example messages.

Further evaluation is necessary to establish whether dynamic routing provides consistent advantages over simpler fusion methods. Such evaluation should include multiple datasets, repeated runs, and more detailed analysis of individual predictions.

9. Discussion and Limitations

9.1 Discussion of Experimental Findings

The experimental results indicate that the proposed Dynamic Evidence Routing model performed strongly on the reported SMS spam detection
test set. Its accuracy of 99.42% and F1-score of 97.64% demonstrate that it correctly classified most of the evaluated messages. The confusion matrix also shows that it produced no false-positive predictions and missed three actual spam messages.

The baseline comparison provides additional context. TF-IDF with Logistic Regression achieved a lower recall of 70.77%, while TF-IDF with Linear SVM achieved a recall of 87.69%. The proposed model achieved a recall of 95.38%. This difference suggests that the proposed configuration detected a larger proportion of the spam messages in the same reported test set.

The results are consistent with the possibility that combining transformer representations with multiple evidence pathways can support SMS classification. However, because only two conventional baselines were evaluated, the findings should be restricted to these specific comparisons. They do not establish universal superiority over all existing models.

9.2 Importance of Evidence Combination

SMS messages vary considerably in length, vocabulary, and intent. Some unwanted messages contain explicit promotional language, while others use more indirect expressions. Legitimate messages can also contain words commonly associated with spam, such as free, offer, or claim, depending on the surrounding context.

A system that considers only one type of evidence may not capture every relevant characteristic. Combining semantic, pattern, and context/entity pathways provides a framework for representing different aspects of a message. The routing mechanism is designed to adjust the relative contribution of those pathways.

The reported message-level examples demonstrate variation in routing weights. Nevertheless, routing weights alone do not establish that the model's decisions are fully interpretable or that each pathway represents a distinct linguistic concept. Additional analysis would be needed to validate the meaning and reliability of the individual pathways.

9.3 Practical Implications

The proposed system may be useful as a research prototype for SMS spam filtering. Its classification output can help distinguish likely spam from legitimate messages, while confidence scores and routing information can provide additional information about individual predictions.

A Flask interface makes the model accessible through a simple message-entry workflow. This demonstrates a possible way to connect the trained classifier to a user-facing application. However, a working interface is not equivalent to a production-ready spam filtering service. Before deployment, the system would require testing under realistic message traffic, monitoring of false-positive and false-negative errors, security checks, performance testing, and procedures for updating the model as spam behaviour changes.

9.4 Dataset Limitations

The reported evaluation uses a test set of 517 messages. Although the results are encouraging, a test set of this size may not represent the full range of messages encountered in practice. Spam campaigns, language use, message formats, and legitimate communication styles can vary across regions, users, and time periods.

The dataset source and detailed composition should be explicitly documented in the final

paper. Without this information, it is difficult to determine how representative the evaluation is of real-world SMS traffic. Future work should evaluate the model on additional datasets containing diverse spam categories, legitimate messages, and messages collected under different conditions. This would provide stronger evidence about generalization beyond the current test set.

9.5 Evaluation Limitations

The reported metrics describe performance on one test set and should not be treated as guaranteed performance on unseen datasets. The observed precision of 100.00% means that no false positives occurred in this evaluation; it does not prove that false positives will never occur. Similarly, the model's 95.38% recall indicates that some spam messages were missed. These errors warrant further investigation because missed spam may still cause inconvenience or expose users to fraudulent content.

Repeated experiments using different random seeds, additional evaluation datasets, and confidence intervals would help assess the stability of the results. Statistical significance testing would also help determine whether the differences between the proposed model and the baselines are larger than would reasonably be expected from sampling variation.

9.6 Ablation Limitations

The ablation study shows that removing the semantic expert or pattern expert changed some classification metrics. However, removing the context/entity expert and replacing dynamic routing with static equal fusion did not change the reported aggregate scores.

These observations do not establish that the affected components are universally useful or unnecessary. Their contributions may depend on the dataset, model training, and types of messages evaluated. In addition, aggregate metrics can remain unchanged even when individual predictions differ.

Future ablation studies should compare predictions at the message level, repeat training under controlled conditions, and test the configurations on more diverse datasets. This would provide a more complete understanding of the contribution of each component.

9.7 Robustness and Generalization

Robustness is important because spam messages can change their wording, introduce unusual spelling, use abbreviations, or attempt to avoid familiar patterns. A model that performs well on ordinary examples may behave differently when confronted with such variations. A preliminary robustness test involving only five examples is insufficient to support a broad claim of robustness. A more reliable evaluation should include a larger collection of deliberately varied messages, including spelling changes, paraphrases, unseen promotional phrases, and legitimate messages containing suspicious-looking words. The system should also be tested on messages from sources different from those used during training. Such tests can help identify distribution

shifts and determine whether model performance deteriorates when the characteristics of incoming messages change.

9.8 Computational and Deployment Considerations The proposed model uses a transformer-based architecture, which generally requires more computational resources than a simple TF-IDF classifier. The actual cost depends on the model configuration, sequence length, hardware, batch size, and inference implementation.

Although the application can produce predictions through Flask, further measurements are

needed before making claims about its speed or scalability. Relevant measurements include average inference latency, memory consumption, throughput, and resource usage under concurrent requests.

A practical deployment should also protect message content, restrict access to model files, validate incoming requests, and avoid retaining sensitive SMS content unnecessarily. These considerations are important when handling personal communication.

9.9 Limitations Summary

The main limitations of the current work are the relatively small reported test set, the need to document the exact dataset and training configuration, the limited number of evaluated baselines, and the absence of sufficient large-scale robustness testing. In addition, the ablation results do not show a measurable aggregate improvement from dynamic routing over static equal fusion on the current test set.

These limitations do not invalidate the reported results. Rather, they define the scope of the conclusions and identify the experiments needed to strengthen the evidence for the proposed method.

10. Conclusion and Future Work

10.1 Conclusion

This research presented an SMS spam detection approach based on a DistilBERT-based model and dynamic evidence routing. The proposed architecture combines semantic, pattern, and context/entity evidence pathways to support binary classification of SMS messages into spam and ham. A Flask-based interface was also used to demonstrate how the trained model can be accessed through a message-entry application. On the reported test set of 517 messages, the proposed model achieved 99.42% accuracy, 100.00% precision, 95.38% recall, and a 97.64% F1-score. Its confusion matrix contained 452 true negatives, 62 true positives, three false negatives, and no false positives.

The model was compared with TF-IDF plus Logistic Regression and TF-IDF plus Linear SVM. The proposed model achieved higher reported recall and F1-score than both baselines on the same test set. These results suggest that the implemented architecture is promising for SMS spam classification under the evaluated conditions.

The ablation study provided additional information about the model's components. Removing the semantic expert reduced recall, while removing the pattern expert reduced precision. Removing the context/entity expert and using static equal fusion produced the same aggregate metrics as the complete model. Therefore, the current experiments support the effectiveness of the overall classifier on the test set, but they do not establish that dynamic routing consistently outperforms simpler fusion strategies.

The research provides a basis for further investigation into evidence combination for SMS spam detection. More extensive evaluation is needed to establish the generalization, robustness, computational efficiency, and practical reliability of the approach.

10.2 Future Work

Several directions can be explored to improve and validate the proposed system.

First, the model should be evaluated on larger and more diverse datasets. This would help determine whether its performance remains stable across different message sources, spam categories, and legitimate communication styles. Dataset documentation should include the source, class distribution, preprocessing procedures, and data-splitting method.

Second, future experiments should compare the proposed model with additional baselines, including a DistilBERT-only classifier, if a properly trained and evaluated version is available.

Comparing the full model with a transformer-only baseline would help determine whether the evidence pathways provide measurable benefits beyond transformer representations alone.

Third, the dynamic routing mechanism should be investigated through repeated experiments and prediction-level analysis. Such experiments should compare dynamic routing with static fusion under controlled conditions and assess whether routing weights consistently correspond to useful differences in message content.

Fourth, the three false-negative predictions should be examined to understand why those spam messages were missed. This error analysis may reveal weaknesses that can be addressed through improved training data, preprocessing, or model design.

Fifth, robustness testing should include larger collections of messages with spelling variations, paraphrasing, unusual formatting, and previously unseen spam expressions. Testing on messages from different sources would also help identify distribution shifts.

Finally, deployment-oriented work should measure inference latency, memory usage, and throughput. Additional security and privacy measures should be implemented before considering the system for real-world use.

Overall, the proposed system demonstrates promising SMS spam classification performance on the reported test set. Further experiments will be essential to determine whether its performance and routing behaviour remain reliable across broader conditions.

Part 5 — References

Use the following references for the background, methodology, transformer model, and implementation tools discussed in your paper. The citations should also appear in the relevant sections of the paper. References

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